

Deviations from the resonance theory of tempo induction

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In: R. Parncutt, A. Kessler & F. Zimmer (Eds.)
Proceedings of the Conference on Interdisciplinary Musicology (CIM04)
Graz/Austria, 15-18 April, 2004 <http://gewi.uni-graz.at/~cim04/>

Background in music psychology. The salience of pulse perception in listeners attending to musical rhythms has been modeled as a band-pass filter (Parncutt 1994) and a simple resonator with a resonance frequency around 2 Hz (van Noorden and Moelants 1999). These models have been able to explain perceptual responses to a variety of rhythmic pulse trains and to excerpts of music from a wide range of styles. In these studies, the models were applied to predict distributions of perceived tempo for large collections of music. However, in many pieces the tempo is ambiguous, allowing several metric levels to serve as the beat. Is this model also able to explain the distribution of responses of groups of listeners to individual excerpts of music?

Background in computation (music information retrieval). Many techniques for the automatic extraction of musical tempo from audio files have been developed, including multiple resonator methods (Scheirer 1998), multiple agent methods (Goto and Muraoka 1998), and probabilistic methods (Cemgil et al. 2001). It is typical for such systems to provide a single tempo value or multiple candidate values, often at different metric levels. However, in view of the inherent ambiguity, a single tempo value does not sufficiently describe the group response, e.g., perceived beat, of a single excerpt of music. A better method would accurately describe the relative ambiguity of tempo across group responses.

Aims. We evaluate the appropriateness of a resonance model for tempo induction to predict the tempo perception of a group of listeners to single excerpts of music and we identify rhythmic and acoustic properties for excerpts that deviate from a resonator-model prediction of their perceived tempo.

Method. Two experiments were done in which subjects ranging in musical training listened to and tapped to the beat of excerpts of music comprising a broad set of styles. Histograms of all listeners' tempos were generated for each excerpt and tempos at the two largest peaks in the histograms were recorded and taken to represent the dominant perceived tempos of that excerpt. The relative strength of these peaks was then compared with the distribution as predicted by the resonance model.

Results. The resonance model predicts a majority of the tempo distributions but in some cases, subjects tapped at metric levels higher or lower than the model would predict. The musical content of the excerpts was examined in order to see what elements in the stimuli create deviations from the model and what elements contribute to the ambiguity of tempo. Deviations from the resonance theory for pulse perception seem to be generally predicted by a statistic designed to represent periodic dynamic accents in the acoustic waveform of the musical excerpt.

Conclusions. The resonance model for pulse perception is applicable to the perception of tempo in a wide variety of cases but fails for some individual musical excerpts. A preliminary description of the relation between musical content and the perception of extreme tempi and tempo ambiguity is given, providing a starting point for more specific research in the field.

The salience of pulse perception in listeners attending to musical rhythms has been modeled as a band-pass filter (Parncutt 1994; Todd 1999) or a simple resonator with a resonance frequency around 2 Hz (van Noorden and Moelants 1999; Moelants 2002). These models have been able to explain perceptual responses to a variety of rhythmic pulse trains and to excerpts of music from a wide range of styles. In these studies, the models were applied to predict distributions of perceived tempo for large collections of music. However, in many pieces the tempo is ambiguous, allowing several metric levels to serve as the beat. Previous studies on pulse perception have shown that listeners can be drawn to different metric levels by perceptual accents on particular beats, including dynamic, durational and melodic accents (Povel and Okkerman 1981; Parncutt 1994; Toiviainen and Snyder 2003). Here we examine how closely the resonance model for pulse perception (van Noorden and Moelants 1999) is able to explain the distribution of responses of groups of listeners to individual excerpts of music. In addition we look specifically at those musical excerpts for which the resonance model does not predict the distribution and attempt to correlate the deviation from the model with measures of perceptual accent.

A good model for the prediction of perceptual tempo across listeners for individual excerpts of music would be a useful tool for automatic tempo extraction. Many techniques for the automatic extraction of musical tempo from audio files have been developed, including multiple resonator methods (Scheirer 1998), multiple agent methods (Allen and Dannenberg 1990; Goto and Muraoka 1998), and probabilistic methods (Cemgil et al. 2001). It is typical for such systems to provide a single tempo value or multiple candidate values, often at different metrical levels. However, in view of the inherent ambiguity, a single tempo value does not sufficiently describe the group response, e.g., perceived beat, of a single excerpt of music. A better method should accurately describe the relative ambiguity of tempo across group responses. Such a method would allow meaningful quantitative evaluation of tempo extractor systems.

Method

Two experiments were performed to assess how closely the distribution of perceived pulse across listeners follows the resonance theory of tempo induction for individual musical excerpts.

Experiment 1

33 subjects were asked to tap to the “beat” of 24 10-sec excerpts of music. Subjects were staff members and students at Philips Research Laboratories and they had an average of 7 years of musical education. Each musical excerpt was played once to give the subject a preview and then played a second time while the subject tapped out the beat. Tap times were recorded via a computer touchpad (Cirque SmartCat) and used to calculate perceived tempo. Subjects were given the chance to repeat the tapping to a given excerpt as many times as necessary until they felt satisfied that their tapping reflected their perception of the beat.

The excerpts were taken primarily from Western popular music; a breakdown of music genres and the corresponding number of excerpts from each are given in Table 1.

Genre	No. Excerpts
Classical	3
Country	3
Dance	2
Hip-Hop	3
Jazz	3
Latin	2
Reggae	2
Rock	3
Soul	3

Table 1. Distribution of music excerpts in Experiment 1 according to music genre.

Experiment 2

24 subjects tapped to the beat of 60 30-second excerpts of music. Subjects were staff members and students from the musicology department at Ghent University. Their average age was 26 and they had an average of 12 years of musical education. Methods were similar to Experiment 1 except subjects tapped on the “space” key of a computer keyboard.

As stimuli for the experiment, 60 pieces were selected in view of maximizing the stylistic differences and the variety of the musical content. An equal number of pieces (6) were taken from 10 stylistic categories: 1. pop music, 2. rock music, 3. heavy metal & punk, 4. afro-American pop music (R&B, soul, reggae, rap, hip-hop), 5. dance (techno, house, drum & bass), 6. classical music, 7. jazz, 8. folk, chanson & cabaret, 9. ethnic & world music, 10. soundtracks, light classical music. Also within these categories variation was maximized by taking pieces with different characters, from different periods, and with different instrumentations. This stylistic differentiation quasi-automatically leads to a large variability in musical content, so we can expect not only different tempi, but also different levels of beat strength and ambiguity.

Analyses

Perceived tempo in beats per minute (BPM) was calculated from the tap times using linear regression after estimating the beat period from the median tap interval. The reliability of the tapping data was assessed by examining how regular it was; 95% confidence intervals for the linear regression model were calculated from subjects' tap intervals and subjects with 95% confidence intervals greater than 10 BPM were excluded from the analysis (3 subjects from Experiment 1 and none from Experiment 2).

For Experiment 1, two groups of subjects

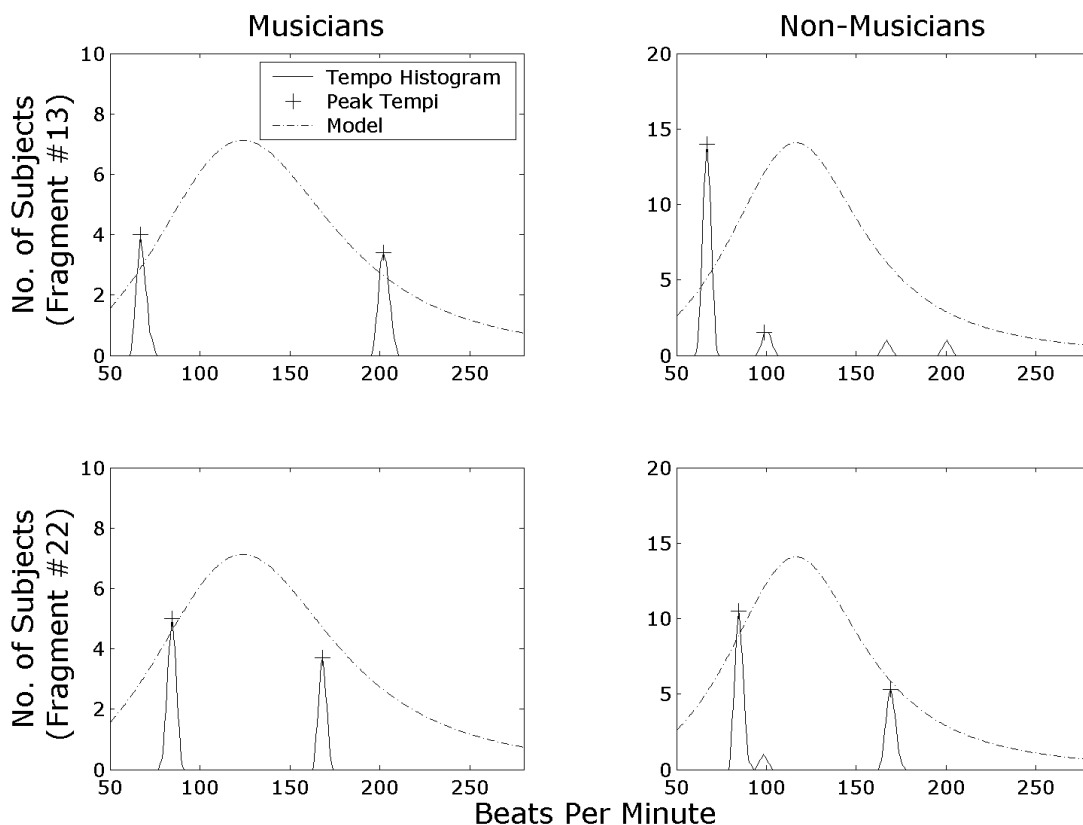
were defined based on their musical background: those with more than 10 years of musical training were called musicians; those with less than 10 years of musical training were called non-musicians. In Experiment 2, all subjects had enough musical education to combine them into one group (musicians).

For each excerpt, a histogram of perceived tempo values was generated and used to estimate the two most dominant perceived tempi. Histograms were generated with binwidths of 0.5 BPM and then smoothed over ten bins. The two highest peaks in the histogram were taken to represent the dominant perceived tempos for that excerpt.

The dominant perceived tempos from all excerpts were used to parameterize a resonance model for pulse perception (van Noorden and Moelants 1999). Data fitting was done using a least-squares method and the following description of the resonance model:

$$R = \frac{1}{\sqrt{(f_o^2 - f^2)^2 + \beta \cdot f^2}} - \frac{1}{\sqrt{f_o^4 + f^4}},$$

where R is the effective resonance amplitude, f_o is the resonant tempo (frequency), β is the damping constant, and f is tempo (frequency). These parameters were fit to a distribution of tapped-tempos for all musical excerpts within each experiment.



Results

Analyses for both Experiment 1 and 2 were the same but the results were somewhat

musicians. In addition, perceived tempi of musicians show a greater deviation from the mean compared to non-musicians, as is reflected in the damping constants, β , 5.0 for

Figure 1. Histograms of tapped tempos for two musical excerpts. Responses from musicians are plotted in the left panels and those from non-musicians are in the right panels. The two most prominent peaks in each histogram are marked (+) and reflect the dominant perceived tempi. The amplitude of the resonance model is shown, using model parameters obtained from fitting the prominent peaks from all histograms to the model. Parameters for the resonance model were obtained separately for musicians and non-musicians. Excerpt #13 (upper panels) is from Frank Morgan's *Mood Indigo* (Jazz); Excerpt #22 is from Bill Withers' *Grandma's Hands* (Soul).

different and are thus presented separately here.

Experiment 1

Figure 1 shows histograms of perceived tempo for two of the 24 musical excerpts. The two most prominent peaks in each histogram are marked and are taken to represent the two most dominant perceived tempi for that excerpt. The histograms are characteristic of those from the rest of the musical excerpts in that they show one or two prominent peaks whose BPM values are related by a simple integer ratio. This relationship reflects the fact that different listeners were tapping at different metrical levels to the same musical excerpt. For Excerpt #13, the non-musicians predominantly tapped at 67 BPM (upper-right panel) while the musicians were split across 67 and 202 BPM (upper-left panel). For Excerpt #22 (bottom panels), both musicians and non-musicians were split across 84 and 168 BPM.

The resonance model amplitude as a function of tempo is also plotted in Figure 1, using model parameters obtained from fitting the model to the dominant perceived tempi of all excerpts across separate groups of musicians and non-musicians (see Method, Fig. 2, and explanation below)

Figure 2 shows scatter plots of the two most prominent peaks from all tempo histograms of musicians (top panel) and non-musicians (bottom panel). The resonance model amplitude, with parameters best fit to the data, is also shown in each case. On average, musicians tended to tap faster than non-musicians, producing a model resonant tempo of 138 BPM compared to 125 BPM for non-

musicians and 2.0 for non-musicians. In other words, the perceived tempo of non-musicians is modeled by a sharper resonance than that for musicians. The resonance model appears to fit the data from musicians better than it does non-musicians, and although correlation coefficients are rather low ($r = 0.7$) the correlations are both significant at the $p < 0.001$ level.

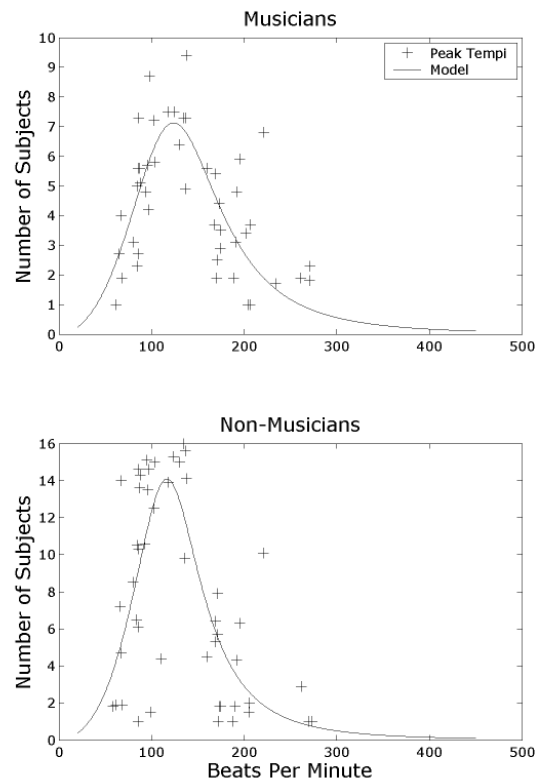


Figure 2. Histogram peaks for data from all musical excerpts in Experiment 1 are plotted along with the resonance amplitude from the model with parameters best fit to the data. For musicians, the best-fit model parameters were $f_o = 138$ BPM, $\beta = 5.0$ and the

correlation coefficient of the fit is $r = 0.71$. For non-musicians, $f_o = 125$ BPM, $\beta = 2.0$ and $r = 0.70$. Correlations are significant at the $p < 0.001$ level for both cases.

Perceived tempo distributions of some individual music excerpts are well explained by the resonance model but others are not. Figure 1 shows that, in response to Excerpt #22 (bottom panels), the distribution of perceived tempi, i.e., peaks of the histogram, falls reasonably close to the model resonance curve for both musicians and non-musicians. For Excerpt #13, musicians' responses are also well predicted by the model (upper left panel), but non-musicians' responses were dominated by the perception of a low tempo at 67 BPM and are not predicted by the resonance model. These deviations from the resonance model are treated in more detail below, along with those from Experiment 2.

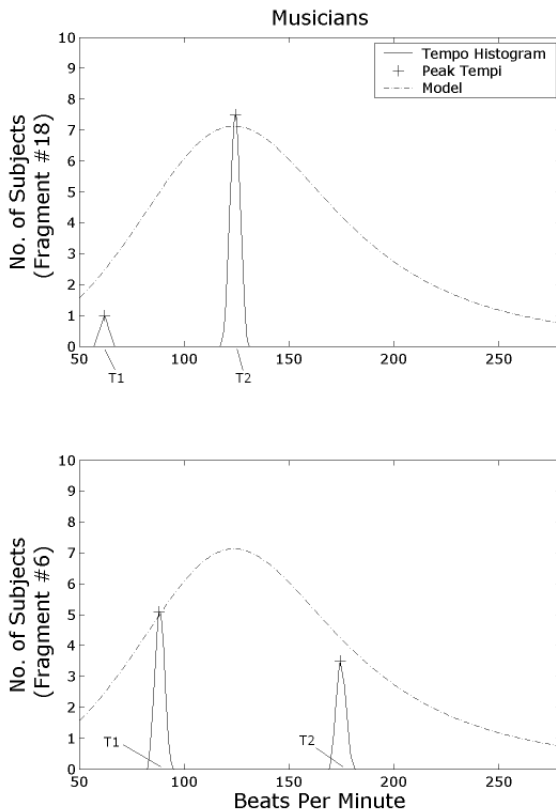


Figure 3. Histograms of tapped tempos from musicians for two musical excerpts (same format as Fig. 1). The lowest and highest dominant tempos are denoted $T1$

and $T2$, respectively. For Excerpt #18 (top panel), the predominant perceived tempo (at $T2$) falls near the model resonant tempo and is relatively unambiguous across subjects. For Excerpt #6 (bottom panel), the predominant perceived tempos straddle the model resonant tempo and the overall perception of tempo is ambiguous across subjects. Excerpt #18 is from Bob Marley's *Jammin'* (Reggae); Excerpt #6 is from Willie Nelson's *No Love At All* (Country).

A consequence of the postulated resonance in tempo perception is that the relation between the predominant perceived tempi and the resonant tempo of the model could be used to predict the ambiguity of tempo across listeners (and vice versa). That is, if a musical excerpt contains a metrical level whose tempo lies near the resonant tempo, the perceived tempo across listeners (i.e., perceived tempo distribution) is likely to be dominated by the tempo of that metrical level and be relatively unambiguous (Fig. 3, upper panel). If, on the other hand, the tempos of adjacent metrical levels straddle the resonant tempo, the perception across listeners would tend to be split evenly across those tempos and be more ambiguous (Fig. 3, lower panel).

To further examine the relationship between tempo ambiguity and the resonance model we calculate a measure of perceived tempo ambiguity, A , from the histogram of tapped tempos as follows:

$$A = \frac{2 \cdot \max(H(T1), H(T2))}{H(T1) + H(T2)},$$

where $H(T1)$ is the height (i.e., number of subjects) of histogram peak at the lowest dominant tempo and $H(T2)$ is the height of the peak highest dominant tempo (see Fig. 3). Thus, the ambiguity, A , can range from 0.0 (in the case where there is only one real peak and the second peak has zero height) to 1.0 (in the case where both peaks are of equal height).

Figure 4 shows measures of tempo ambiguity, A , for each music excerpt plotted as a function of the mean of the two dominant tempos from each histogram. The resonant theory holds that the ambiguity measure should be low for mean tempi that fall below or above the resonant tempo and large for

mean tempi that fall near the resonant tempo. This is generally seen in the data but is more apparent in the responses from the musicians (upper panel). An analysis of variance on two groups of ambiguity measures, one for mean tempi near the resonant tempo (>110 and <170 BPM) and the other for mean tempi outside this range, yields a significant difference in the means of the groups. This result indicates that the data generally adheres to the resonance theory of tempo perception in terms of the relative ambiguity of tempi across listeners.

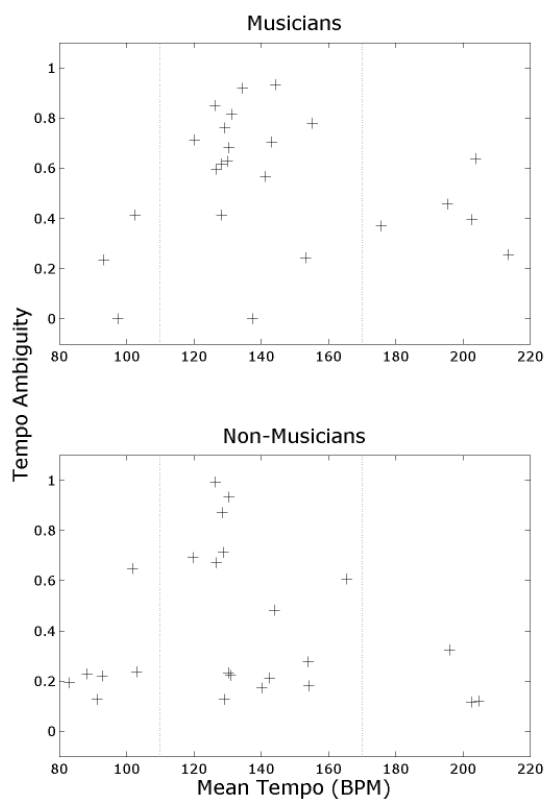


Figure 4. Tempo ambiguity as a function of the mean of the two primary tempos in each tapped-tempo histogram (Experiment 1). Ambiguity was calculated as described in the text. Vertical dashed lines indicate boundaries for separating data into two groups, those with mean tempi near the resonance and those with tempi more distant from the resonance. ANOVA results show significant differences between the mean ambiguity values of the two groups ($p < 0.01$ for musicians, $p < 0.05$ for non-musicians).

To investigate the difference in tempo ambiguity between musicians and non-musicians, we plot calculated ambiguity values from non-musicians against those from musicians for each musical excerpt in Fig. 5. For many excerpts, the ambiguity is relatively the same for musicians and non-musicians. However there is a group of excerpts for which the ambiguity level is quite high for musicians and very low for non-musicians. In all of these cases, non-musicians tended to tap at slower tempos than did the musicians. This is in direct conflict with results of Drake, Penel, and Bigand (2000), who found that musicians tended to tap more slowly than non-musicians, presumably because they were more experienced at following music along at higher metrical levels. In the majority of the cases here in which non-musicians tapped slower than musicians, the excerpt contained a strong baseline, indicating the possibility that non-musicians' perception of tempo could be dominated by heavy activity in the baseline. Further investigation would be required to confirm this qualitative observation.

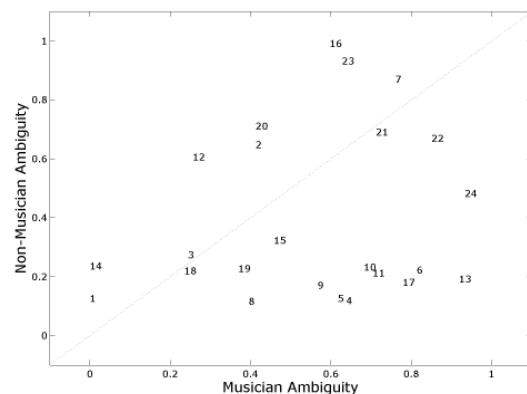


Figure 5. Ambiguity values for non-musicians plotted as a function of that from musicians for each musical excerpt (Experiment 1). Numbers in the plot indicate the excerpt number.

Experiment 2

All subjects from Experiment 2 were assembled into one group (musicians). A scatter plot of the tapped-tempo responses along with the best-fit resonance model is shown in the top panel of Fig. 6. The best-fit model has a resonant tempo of 140 BPM,

similar to that for musicians in Experiment 1. In general, the resonance model does not fit these data as well as it did in Experiment 1, yielding a correlation coefficient of 0.54; the correlation between model and data is not significant.

The bottom panel of Figure 6 shows tempo ambiguity plotted as a function of the mean tempo from the two primary tempos of each musical excerpt (same format as Fig. 4). In this experiment, there is no clear relationship between the mean tempo and tempo ambiguity, i.e., mean tempi near the resonant tempo do not, on average, show a higher ambiguity. This is consistent with the fact that the resonance model does not fit the primary tempi in Experiment 2 as well as it did in Experiment 1.

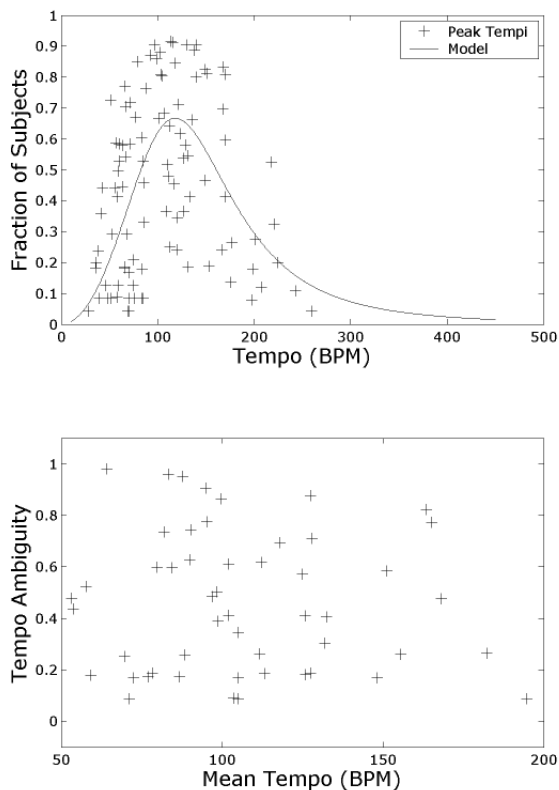


Figure 6. Summary of data from Experiment 2. Top panel: Histogram peaks for data from all musical excerpts are plotted along with the model resonance amplitude, R , with parameters best fit to the data. The best-fit model parameters were $f_o = 140$ BPM, $\beta = 10.0$, and the correlation coefficient of the fit is $r = 0.54$ (not significant). Bottom panel: Tempo ambiguity as a

function of the mean of the two primary tempos in each tapped-tempo histogram. No clear trend is seen in the data.

An example of a musical excerpt whose global tempo response did not fit the resonance model very well is shown in Figure 7, *Phaéton* from Jean-Baptiste Lully. While the piece is played at a tempo close to the preferred/resonant tempo (130 BPM at the quarter-note level), the majority of subjects tapped at half this rate. The tapped-tempo histogram is shown in Fig. 8 along with the best-fit resonant model output. A possible explanation for this deviation is that the first and third beats of measures throughout the excerpt receive perceptual accents, drawing listeners to tap at those beats. We know from previous studies that there are a number of sources for perceptual accents in rhythmic sequences, including dynamic accents, durational accents (longer events are perceived to be accented in comparison to shorter events), and melodic accents (Povel and Okkerman, 1981; Parncutt, 1994; Toiviainen and Snyder, 2003).



Figure 7. Excerpt from Jean-Baptiste Lully's *Phaëton*. Dotted quarter notes and accents serve to provide durational and dynamic accents on the first and third quarter-note beats of most measures. The tempo of the recorded excerpt was 130 BPM at the quarter-note level.

Examination of the score shown in Fig. 7 shows that the quarter notes on beats one and three of many measures are dotted and accented and thus could contribute to durational and dynamic accents on those beats. The accentuation is enhanced in the performance by 'overdotting', a performance style typical for French baroque music and especially for French overtures: dotted figures are played as if they were double-dotted, thus creating a very short upbeat to the main beat. It is possible that these accents encourage subjects to tap on beats one and three and not on two and four.

Acoustic cues to resonance deviation

In order to test the idea that perceptual accents in the musical excerpts are indeed the cause for deviations of perceived tempo from the resonance theory of pulse perception, we tried to predict, from acoustic analyses, which excerpts would elicit deviant tempo responses. The most direct accent type to gauge via acoustic analysis is the dynamic accent, which we assume is related to the relative energy in a time-limited portion of the signal.

The first step in this analysis is to define a value to represent the deviation of perceived tempo from the resonance model. For this, we used the two dominant tempos from each tapped-tempo histogram, $T1$ and $T2$, and simply defined the resonance deviation, Δ_r , to be:

$$\Delta_r = ((R(T1) - H(T1)) + (H(T2) - R(T2))),$$

where $R(T1)$ and $R(T2)$ are the resonance model amplitudes at tempos $T1$ and $T2$, and $H(T1)$ and $H(T2)$ are the histogram values at $T1$ and $T2$. A negative value for Δ_r indicates that more subjects tapped to a lower tempo than the resonance model would predict and a positive value indicates the opposite.

Next we calculated a statistic, Δ_{ap} (acoustic periodicity difference), from the acoustic waveform of each musical excerpt to represent the relative strengths of periodicities at tempos $T1$ and $T2$. Such a statistic should embody the relative strength of dynamic accents occurring at time intervals corresponding to $T1$ and $T2$ and therefore allow us to relate acoustic dynamic accents to Δ_r , our measure of deviation from the resonance model.

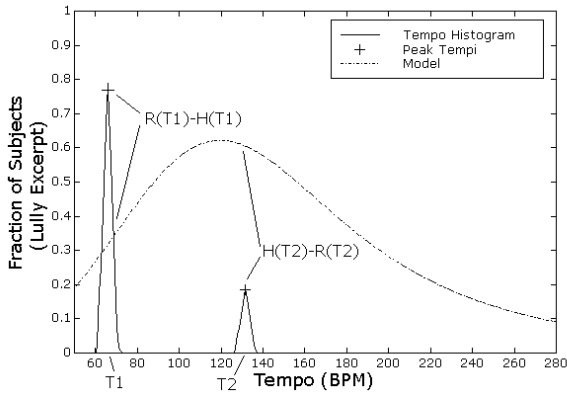


Figure 8. Histogram of tapped response tempos to excerpt from J.-B. Lully's *Phaëton* (Fig. 7). The majority of the subjects tapped near 65 BPM (T1) despite the fact that the next higher metrical level falls close to the tapping resonance (130 BPM, T2). Our method for determining the quantitative deviation of the histogram (H) from the resonance model (R) is illustrated: $\Delta_r = (R(T1) - H(T1)) + (H(T2) - R(T2))$. Both components are negative in this case.

The acoustic periodicity difference, Δ_{ap} , was calculated in the following manner:

- The excerpt was band-pass filtered into 6 frequency bands: 0-200, 200-400, 400-800, 800-1600, 1600-3200, and 3200-22050 Hz (further processing was the same in each band and occurred independently).
- The instantaneous power of the signal was calculated, low-pass filtered, and down-sampled to 500 Hz in order to estimate the temporal envelope in each band.
- Period histograms, $P_k(T1)$ and $P_k(T2)$, (where k is the frequency-band number, 1-6) were generated from the temporal envelopes at periods corresponding to the tempos $T1$ and $T2$.
- The energy, $E_k(T1)$ and $E_k(T2)$, in the period histograms were calculated by summing each period histogram and dividing by the length of the excerpt (number of samples).

- The synchronization index¹, $S_k(T1)$ and $S_k(T2)$, was calculated from each period histogram to estimate the strength of periodicity at tempos $T1$ and $T2$:

$$S_k(T) = \left| \frac{\sum_{n=1}^N P_k(T) \cdot e^{\frac{-j2\pi n}{N}}}{\sum_{n=1}^N P_k(T)} \right|,$$

where N is the length of the period histogram.

- Finally, the energy values and the synchronization indices were combined across frequency bands to form the acoustic periodicity difference:

$$\Delta_{ap} = \sum_{k=1}^6 E_k(T2) \cdot S_k(T2) - E_k(T1) \cdot S_k(T1).$$

The value of Δ_{ap} is positive if there is more energy, i.e., dynamic accents, in the musical excerpt at the periodicity defined by $T2$ than that by $T1$. It is negative if the reverse is true. Our working hypothesis is that this measure of dynamic accent should correlate positively with Δ_r , i.e., the likelihood that subjects tap at tempo $T2$ rather than $T1$.

The resonance deviation, Δ_r , is plotted against the acoustic periodicity difference, Δ_{ap} , for both experiments in Fig. 9. There appears to be no obvious relationship between the variables in Experiment 1, but there is a significant positive correlation ($r = 0.56$, $p < 0.001$) for data from Experiment 2 (lower panel). The effect appears to be weaker for small deviations but is quite strong for data points at large values of Δ_r and Δ_{ap} . The data point for the Lully excerpt (Figs. 7 and 8) is circled in the lower panel of Fig. 9: the leftward shift in the tapped-tempo histogram ($\Delta_r = -0.88$) is associated with a relatively large amount of periodic energy at tempo $T1$ compared to that at tempo $T2$ ($\Delta_{ap} = -0.003$).

A possible explanation for the weak correlation between Δ_r and Δ_{ap} in Experiment 1 is that the responses, on average, show smaller deviations than those in Experiment 2. When comparing the upper and lower panels of Fig. 9 one sees that the variance of

both variables is much less for Experiment 1 than it is for Experiment 2.

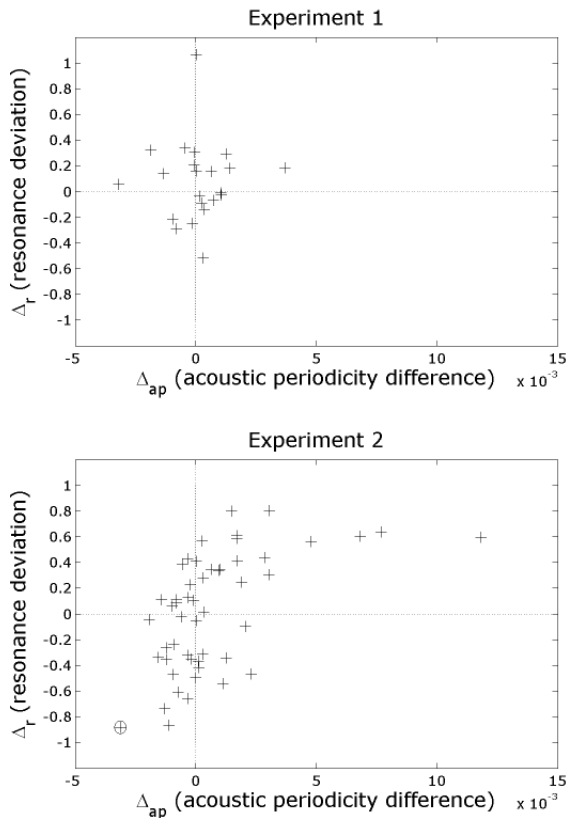


Figure 9. Resonance deviations plotted against the acoustic periodicity difference for each musical excerpt. The circled data point in the bottom panel is from responses to and calculations on the Lully excerpt shown in Fig. 7.

Thus, our measure of dynamic accent periodicity does not fully account for deviations from the resonance model for pulse perception but it does seem to be a good indicator for the largest deviations. Perhaps it could be combined with acoustic measures of durational or melodic accents to more completely explain the data. It is less straightforward, however to extract accurate measures of these melodic or durational accents from a polyphonic music signal than it is for dynamic accents.

Conclusions

The resonance model for pulse perception is applicable to the perception of tempo in a wide variety of cases but fails for some individual musical excerpts. A preliminary description of the relation between musical content and the perception of extreme tempi and tempo ambiguity are given, providing a starting point for more specific research in the field. A measure of acoustic dynamic accent periodicity is able to account for large deviations in responses from the resonance model but does not fully explain smaller deviations. Measures of other accent types, such as durational or melodic accents, could help further explain the small deviations. The relationships between these acoustic measures and the ambiguity of tempo, as measured across groups of listeners, are likely to be important statistics for automatic tempo extraction systems.

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¹ The synchronization index is often used in neuroscience to measure the degree of phase-locking in a neural response in response to a periodic stimulus (Johnson, 1980).